

Replication of Study X by Sample & Sample (20xx, Psychological Science)

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Introduction

[No abstract is needed.] Each replication project will have a straightforward, no frills report of the study and results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of data collection. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for replication, and any other essential information. You will likely contact the authors of the

original study, so this should be professional and forward facing. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

This paper developed a behavioral paradigm used to measure the capability of facial recognition in two closely related species of paper wasp. They demonstrated a previously undocumented form of higher social cognition in an invertebrate, and used a species comparison to shed light on the potential evolutionary trajectory of this behavior. These concepts are pertinent to fields of animal social behavior and evolutionary neurobiology, which I am very interested in.

I will need to simulate the raw learning curve data of the wasps for each trial and in each condition. I plan to do a 2x3 study (2 species of wasp x 3 stimuli conditions of face, caterpillar, and pattern) with n=480 trials in each condition (12 wasps with 40 trials each). This is a total of 2880 trials across all conditions with 72 simulated wasps (36 from each species). In order to simulate learning data, I need to be able to increase the likelihood of a correct trial for each wasp at each iteration. The difficult part in doing this simulation will be figuring out how to capture the individual variation in learning ability for each wasp, which will be affected by its species and condition placement. After I have this data, I will need to use generalized estimating equations (GEE) to analyze learning curves with respect to individual wasp differences. Another difficulty will be in understanding how to implement this model and interpret the results.

[link to project repo](#)

[link to original paper in repo](#)

[link to original paper publishing site](#)

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or “none”.

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.